

CLEF 2026 SimpleText • Task 1.1 & Task 1.2 • Biomedical Domain

NUQLab at SimpleText 2026: Plan-Guided Biomedical Scientific Text Simplification

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38.96

SARI — Task 1.1
official CLEF 2026 evaluation

38.97

SARI — Task 1.2
official CLEF 2026 evaluation

No fine-tuning

inference only, from public
Cochrane-auto checkpoints

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The Task and the Challenge

CLEF 2026 SimpleText Task 1 — simplify scientific text, and nothing more

Task 1.1 — Sentence level

- Rewrite individual biomedical sentences in a more accessible form while preserving meaning
- Evaluated on the Cochrane-auto biomedical corpus

Task 1.2 — Document level

- Simplify a full source document, not isolated sentences
- Requires sentence segmentation and document recomposition

The central challenge: controlled simplification

- Systems that generate freely tend to introduce unsupported or overly creative content — the issue highlighted by the CLEF 2025 hallucination track (Task 2).
- Biomedical abstracts are dense: technical terminology, statistical expressions, and highly compressed discourse.
- Our response: a plan-guided approach — a sentence-level classifier first predicts a simplification operation, then the generator is conditioned on the predicted control token.
- This makes simplification more structured and interpretable than unconstrained sequence-to-sequence generation.

Main Contributions

What this submission adds

1

A unified plan-guided inference pipeline

One classify-then-generate design serving both sentence-level and document-level biomedical simplification.

2

Adaptation of Cochrane-auto checkpoints

Released pretrained checkpoints transferred to CLEF 2026 SimpleText Task 1 without any further training.

3

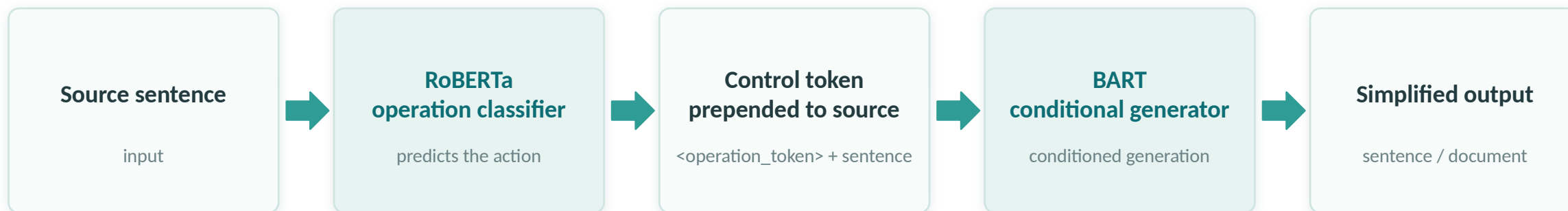
Practical inference-time robustness mechanisms

Delete handling, a tokenizer-based control-token diagnostic, and biomedical-aware document splitting.

The system is intentionally lightweight: it reuses released pretrained Cochrane-auto models, so the whole pipeline is reproducible from public checkpoints.

System Architecture

Two-stage plan-guided pipeline — all models run in inference mode only; no weights are updated



Label mapping — generic classifier outputs are mapped manually to the operation space

LABEL_0 → *ignore*

LABEL_1 → *rephrase*

LABEL_2 → *split*

LABEL_3 → *merge*

LABEL_4 → *delete*

The mapping is verified at load time using validation-set label-distribution inspection.

Soft conditioning, not a hard constraint. The control token conditions the generator rather than constraining decoding, so the output may simplify a sentence without fully realising the predicted structural operation.

Pretrained Models Used

All checkpoints are public and used in inference mode only — no additional fine-tuning

Role	Checkpoint	Architecture
Operation classifier	janbakker/classifier-cochraneauto	RoBERTa
Plan-guided generator	janbakker/o-bartsent-cochraneauto	BART
Fallback generator	janbakker/bartsent-cochraneauto	BART

Why reuse instead of retrain?

- The pipeline is reproducible from public checkpoints alone
- No training cost, no external API calls
- Behaviour is fully inspectable at inference time

Implementation

- Single NVIDIA T4 GPU in Google Colab
- PyTorch and Hugging Face Transformers
- Random seed set to 42

Plan-Guided Generation: Robustness at Inference Time

Four mechanisms that keep the pipeline from failing silently

1 Control-token detection

Tokens are detected automatically from the generator tokenizer, checking both plain forms (e.g. rephrase) and bracketed forms (e.g. <rephrase>).

2 Load-time diagnostic

A check verifies that distinct operations resolve to distinct tokens. If all operations collapse to one token, plan-guided mode is disabled and the system falls back to the plain BARTsent generator.

3 Delete and degenerate handling

Delete-predicted sentences are assigned an empty string and excluded from generation. Generated outputs of three words or fewer are treated as degenerate and replaced by the original source sentence.

4 Non-English inputs

Copied through unchanged, since the released checkpoints are English biomedical models — this avoids noisy cross-lingual predictions.

The plain BARTsent configuration is retained throughout as a secondary fallback.

Document-Level Extension (Task 1.2)

Same classify-generate backbone, plus biomedical-aware segmentation and recomposition

Biomedical-aware sentence splitter

Built on NLTK Punkt. Before tokenization, the splitter protects patterns frequently mis-segmented in biomedical text:

- p-values and confidence intervals
- statistical abbreviations
- common biomedical abbreviations, and decimal numbers

Placeholders are restored after tokenization. Very short fragments are merged into the preceding sentence; segments shorter than two words are discarded. If NLTK fails, the splitter falls back to a regex-based routine.

Recomposition

- Each sentence goes through the same classify-generate pipeline as Task 1.1
- Simplified outputs are concatenated with a single space to form the final document
- Delete-predicted sentences are omitted from the recomposed document

The DELETE_OUTPUT switch — handling the all-delete document

source (used for the submitted Task 1.2 run): falls back to the first source sentence, so the submission contains no empty document-level predictions.

empty (used for the Task 1.1 submission): emits an empty string for such cases.

Experimental Setup

Test data and inference settings

Task 1.1 data

- Test file: 48,809 sentences from 1,734 documents
- 20,259 English (41.5%) · 28,550 non-English (58.5%)
- Primary evaluation subset: English Cochrane-auto 2026 portion, 3,679 sentences

Task 1.2 data

- Test file: 8,069 documents
- Primary evaluation subset: 175 English Cochrane-auto documents
- Validation split: 119 documents

Inference settings

Setting	Task 1.1	Task 1.2
Classifier batch size	64	64
Classifier max length	128	128
Generator batch size	32	32
Generator max input length	256	256
Generator max new tokens	150	128
Beam width	8	5
Length penalty	1.5	1.0
Degenerate fallback	source if ≤ 3 words	

Results — Task 1.1 (Sentence Level)

SARI is used for offline validation, following the task setup

System	Evaluation setting	SARI
Plain BARTsent (no control tokens)	Offline validation	≈ 30–32
Plan-guided (earlier run)	Offline validation	≈ 35.5
Plan-guided (this system)	Offline validation	51.96
Plan-guided (this system)	Official leaderboard	38.96

Offline validation is computed on the non-delete Cochrane-auto validation rows (rows with non-empty reference simplifications); the official score is the CLEF 2026 leaderboard result on the full multilingual test set. Degenerate generations of three words or fewer are replaced by the source sentence to preserve the keep component.

51.96

offline validation SARI
(non-delete rows)

38.96

official CLEF 2026
leaderboard SARI

13.9%

of the 48,809 submitted rows
are empty (delete-predicted)

Results — Task 1.2 and Why the Design Works

Document-level performance is essentially on par with the sentence-level result

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SARI — Task 1.2, official CLEF 2026 evaluation

- Same plan-guided classify-generate backbone, extended with the biomedical-aware splitter and single-space recomposition
- The submitted run uses the source delete-handling setting, so fully-deleted documents fall back to their first source sentence — the submission contains no empty document-level predictions
- Offline validation over the 119 Cochrane-auto validation documents was used during development to confirm the pipeline before submission

Three design choices account for the result

1 Explicit operation prediction

Predicting the operation before generation makes simplification more structured than unconstrained sequence-to-sequence generation.

2 Control-token diagnostic

Prevents silent plan-guided failures by reverting to plain BART when the tokenizer cannot distinguish operations.

3 English-only restriction

Prevents the English biomedical model from being applied to non-English inputs, avoiding noisy cross-lingual predictions.

Caveat: delete predictions are common. While this increases brevity, overuse of deletion can reduce informativeness — the plain BARTsent configuration is provided to investigate this trade-off in future runs.

Plan Adherence and the Case for a Lightweight Pipeline

Two issues that matter for interpreting the system's behaviour

1. Is the predicted plan actually followed?

- The control token conditions the generator rather than constraining its decoding, so plan adherence is not architecturally guaranteed
- The generator can produce a fluent simplification while only partially realizing the requested operation
- This is most likely for the structurally demanding split and merge labels, which require reorganizing sentence boundaries rather than substituting or dropping content
- We did not run a systematic manual audit for this submission — we report this as an expected consequence of soft conditioning, not as a measured error rate
- Next step: a labeled audit of a sample of Task 1.1 and 1.2 outputs, checking whether the realized operation matches the predicted one

2. Why not just use a larger LLM?

- The case does not rest on output fluency — larger models are likely to have an advantage there
- It rests on reproducibility and cost: the system runs entirely from public checkpoints on a single T4 GPU
- No additional training and no external API calls
- Behaviour — including the delete rate and control-token resolution — is fully inspectable at inference time
- A useful low-resource baseline where reproducibility and local inference matter, even if a larger LLM might improve fluency or close part of the remaining evaluation gap

Conclusion and Future Work

NUQLab at CLEF 2026 SimpleText Task 1

What we showed

- Pretrained Cochrane-auto plan-guided checkpoints transfer effectively to the CLEF 2026 benchmark without any fine-tuning — a RoBERTa operation classifier combined with a BART generator conditioned on predicted control tokens.
- The key enablers are careful control-token handling, a tokenizer-based failure diagnostic, delete handling with source-sentence fallback, and biomedical-aware sentence segmentation.
- Structured operation prediction is a useful and reproducible way to guide biomedical simplification without additional training.

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SARI — Task 1.1 (official)

38.97

SARI — Task 1.2 (official)

51.96

SARI — non-delete validation subset

Future work: stronger multilingual support · a systematic audit of plan adherence · more faithful document-level control mechanisms beyond sentence-wise recomposition

Thank you

Questions are very welcome.

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